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Seat No.	
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[5668]-203

S.E. (Information Technology) (I Semester) EXAMINATION, 2019
DIGITAL ELECTRONICS AND LOGIC DESIGN
(2015 PATTERN)

Time : Two Hours

Maximum Marks : 50

- N.B. :—**
- (i) Answer Q. Nos. 1 or 2, 3 or 4, 5 or 6, 7 or 8.
 - (ii) Neat diagrams must be drawn wherever necessary.
 - (ii) Figures to the right indicate full marks.
 - (iii) Assume suitable data, if necessary.

1. (a) Convert the following octal number into its equivalent Binary
Decimal and Hexadecimal (357.3)₈. [6]

- (b) Design and draw four bit Excess-3 adder using IC 7483. [6]

Or

2. (a) What is Logic Family ? Explain the terms : [6]

(i) Fan out

(ii) Propagation Delay.

- (b) Design Full Adder using IC 74153. [6]

3. (a) Compare combinational circuits with sequential circuits. Convert
JK Flip-Flop into SR flip-flop. [6]

- (b) Draw 3-bit Ring and Twisted ring counter. Draw state diagram
for 3-bit Ring and Twisted ring counter, assuming initial state
as 001. [7]

Or

4. (a) Design 3-bit Synchronous up counter with JK flip-flops. [6]

- (b) Design a sequence generator to generate the following sequence
10101 using JK flip-flop. [7]

P.T.O.

5. (a) Explain the difference between CPLD and FPGA. [6]
(b) Draw ASM chart for 2-bit binary down counter having one enable line such that : $E = 1$ (Counting enabled), $E = 0$ (Counting disabled). [6]

Or

6. (a) Implement the following function using PLA : [6]
 $F_1(A, B, C) = \Sigma m(0, 2, 5, 6)$ $F_2(A, B, C) = \Sigma m(1, 4, 5, 6)$.
(b) Design and implement 3-bit gray to binary code converter using PAL. [6]
7. (a) State and explain any *three* data types supported by VHDL. [6]
(b) Explain the process statement in behavior model of VHDL with respect to syntax, sensitivity list and declarative part. [7]

Or

8. (a) What is Structural Modeling ? Implement full Adder using Structural Modeling. [6]
(b) What is VHDL ? Explain entity-architecture declaration for 2-bit NAND and OR gate. [7]